

1 Rational Numbers

WARM UP

1. State whether the following are positive or negative rational numbers.
   a. $\frac{-5}{18}$ b. $\frac{9}{-17}$ c. $\frac{128}{137}$ d. $\frac{-145}{-147}$
2. Reduce the following rational numbers in the standard form.
   a. $\frac{364}{-228}$ b. $\frac{11}{154}$ c. $\frac{-56}{-96}$ d. $\frac{121}{-132}$
3. Represent the following rational numbers on a number line.
   a. $\frac{-8}{3}$ b. $\frac{7}{13}$ c. $\frac{-6}{17}$ d. $\frac{19}{11}$
4. Find the absolute value of each of the following.
   a. $\frac{15}{19}$ b. $\frac{-8}{-17}$ c. $\frac{11}{-18}$ d. $\frac{-23}{33}$
5. Which of the following pairs are equivalent rational numbers?
   a. $\frac{-5}{6}, \frac{-35}{42}$ b. $\frac{18}{9}, \frac{-8}{4}$ c. $\frac{11}{16}, \frac{22}{32}$ d. $\frac{-5}{-2}, \frac{50}{20}$
6. Write four equivalent rational numbers for each of the following.
   a. $\frac{-8}{9}$ b. $\frac{3}{5}$ c. $\frac{-7}{15}$ d. $\frac{-9}{-10}$
7. Fill in the blanks with <, > or = sign.
   a. $\frac{16}{7}$ _____ $\frac{-9}{7}$ b. $\frac{-11}{12}$ _____ $\frac{3}{4}$ c. $\frac{15}{20}$ _____ $\frac{3}{4}$ d. $\frac{-8}{-32}$ _____ $\frac{16}{14}$
8. Arrange the following rational numbers in ascending order.
   a. $\frac{6}{8}, \frac{11}{4}, \frac{-13}{16}, \frac{9}{2}$ b. $\frac{-9}{11}, \frac{7}{3}, \frac{13}{6}, \frac{-8}{2}$ c. $\frac{-7}{8}, \frac{6}{5}, \frac{13}{18}, \frac{-1}{3}$ d. $\frac{-5}{7}, \frac{11}{21}, \frac{-8}{3}, \frac{4}{21}$
9. Arrange the following rational numbers in descending order.
   a. $\frac{6}{7}, \frac{-9}{14}, \frac{-8}{28}, \frac{-11}{14}$ b. $\frac{4}{5}, \frac{26}{3}, \frac{-8}{15}, \frac{7}{9}$ c. $\frac{7}{10}, \frac{3}{5}, \frac{-1}{10}, \frac{11}{30}$ d. $\frac{5}{6}, \frac{-1}{4}, \frac{-3}{2}, \frac{13}{-14}$
10. Write 10 rational numbers between -5 and -3.
11. Write 5 rational numbers between $\frac{5}{7}$ and $\frac{7}{9}$.
12. Subtract the sum of $\frac{5}{7}$ and $\frac{3}{14}$ from the sum of $\frac{11}{3}$ and $\frac{6}{7}$.

RATIONAL NUMBERS

A **rational number** is a number that can be written in the form of $\frac{p}{q}$, where p and q are integers, and $q \neq 0$. It

includes all integers and all fractions with positive and negative signs. Some examples of rational numbers are $-9, \frac{8}{3}, 0, 63\%, 5.64, \sqrt{36}, \sqrt[3]{-8}$ and -7.

Note here that the numbers of the form $\frac{-7}{0}$ or $\frac{4}{0}$ cannot be defined and hence are not included in the set of rational numbers.

Remember

- Every natural number, say p can be written as $\frac{p}{1}$, which is a rational number, but a rational number such as $\frac{1}{p}$ need not be a natural number.
- Zero is a rational number, as it can be expressed in the form of $\frac{p}{q}$ as $\frac{0}{q}$, where $p=q$ and q is any non-zero integer.
- Every fraction is a rational number, but a rational number need not be a fraction. For example $\frac{-7}{5}$ is a rational number but not a fraction.

In this chapter, we will review the basics of rational numbers, four fundamental operations on rational numbers and the properties of rational numbers on these operations.

REPRESENTATION OF RATIONAL NUMBERS ON THE NUMBER LINE

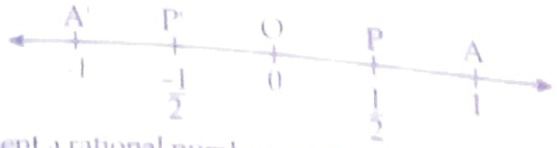
We know that the positive rational numbers lie to the right side of 0 and negative rational numbers lie to the left side of 0 on the number line.

Draw a number line and mark point 0 on it. Now, mark points –1 and 1 at equal distances on opposite sides of 0. Label the point 0 as O, 1 as A and –1 as A'.



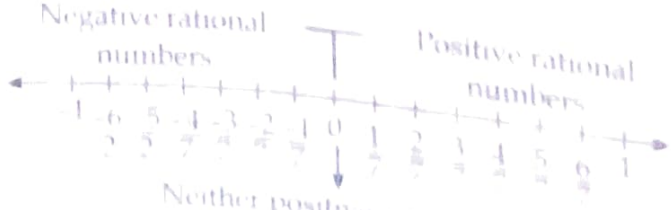
Now, divide each unit length, i.e., 0 to 1 and –1 to 0 into two equal parts. Let P be the midpoint of OA and P' the midpoint of OA'. The lengths of OP and OP' are half the lengths of OA and OA', respectively.

Thus, the points P and P' represent $\frac{1}{2}$ and $\frac{-1}{2}$, respectively.



In general, we can say that to represent a rational number on the number line, divide each unit length on the number line into as many parts as the denominator of the rational number and move as many steps starting from O on the number line, as the numerator (towards the right for positive rational numbers and towards the left for negative rational numbers). We will get the required rational number.

Look at the following number line:



Here, each unit length is divided into 7 equal parts.

ABSOLUTE VALUE OF A RATIONAL NUMBER

The absolute value of a rational number is a rational number without its sign.

For example,

EQUIVALENT RATIONAL NUMBERS

The equivalent rational numbers are rational numbers that have the same value but are represented differently.

For example, $\frac{1}{3}$, $\frac{2}{6}$ and $\frac{3}{9}$ are equivalent rational numbers as $\frac{1}{3} = \frac{2}{6} = \frac{3}{9}$, i.e., $\frac{1 \times 2}{3 \times 2} = \frac{2}{6}$, $\frac{1 \times 3}{3 \times 3} = \frac{3}{9}$

Finding Equivalent Rational Numbers

We can find equivalent rational numbers by multiplying or dividing the numerator and denominator of the given rational number by the same non-zero numbers, i.e.,

$\frac{a}{b}$ and $\frac{c}{d}$ are equivalent, if $\frac{a \times x}{b \times x} = \frac{c}{d}$ or $\frac{a \div x}{b \div x} = \frac{c}{d}$

For example, a. $\frac{2 \times 3}{5 \times 3} = \frac{6}{15}$, $\frac{2}{3}$ and $\frac{6}{15}$ are equivalent rational numbers.

b. $\frac{21 \div 3}{27 \div 3} = \frac{7}{9}$, $\frac{21}{27}$ and $\frac{7}{9}$ are equivalent rational numbers.

Remember

- If $\frac{p}{q}$ and $\frac{r}{s}$ are two rational numbers, then $p \times s = q \times r$.
- If $p = 0$, then $\frac{p}{q}$ will also be 0.

STANDARD FORM OF RATIONAL NUMBERS

A rational number $\frac{p}{q}$ is said to be in the standard form, if:

- the denominator q of $\frac{p}{q}$ is positive.
- p and q have no common factor, other than 1. In other words, $\frac{p}{q}$ is said to be in the standard form, if p and q are co-prime numbers.

  For example $\frac{12}{13}$ and $\frac{-17}{19}$ are in the standard form as their denominators are positive and are co-prime.
- If the denominator of the rational number is a negative integer, then multiply the numerator and the denominator by -1 to make the denominator positive.

  For example, $\frac{6}{-7} = \frac{6 \times (-1)}{-7 \times (-1)} = \frac{-6}{7}$

Two Methods to Write a Rational Number in the Standard Form

Method 1: Divide the numerator and denominator of the given rational number by their common factors until the numerator and denominator have no common factor, other than 1 (co-prime).

For example, write $\frac{36}{-48}$ in the standard form.

$\frac{36}{-48} = \frac{36 \div 2}{-48 \div 2} = \frac{18}{-24} = \frac{18 \div 2}{-24 \div 2} = \frac{9}{-12} = \frac{9 \div 3}{-12 \div 3} = \frac{3}{-4} = \frac{3 \times (-1)}{-4 \times (-1)} = \frac{-3}{4}$

So, $\frac{-3}{4}$ is the standard form of $\frac{-36}{48}$.

Method 2: Divide the numerator and denominator of the rational number by the HCF of its numerator and denominator.

For example, write $\frac{72}{81}$ in the standard form.

$\frac{72}{81} = \frac{72 \div 9}{81 \div 9} = \frac{8}{9}$ (HCF of 72 and 81 = 9). So, $\frac{8}{9}$ is the standard form of $\frac{72}{81}$.

Solved Examples

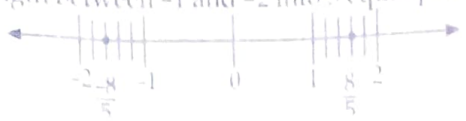
Example 1: Represent $\frac{8}{5}$ and $\frac{-8}{5}$ on the number line.

Solution: We know that $\frac{8}{5} = 1\frac{3}{5} = 1 + \frac{3}{5}$ and $\frac{-8}{5} = -1\frac{3}{5}$.

Thus $\frac{8}{5}$ is greater than 1 and lies between 1 and 2.

Similarly $\frac{-8}{5}$ is less than −1 and lies between −1 and −2.

To represent $\frac{8}{5}$ on the number line, we divide the unit length between 1 and 2 into 5 equal parts and move 3 steps towards the right. Similarly, to represent $\frac{-8}{5}$ on the number line, we divide the unit length between −1 and −2 into 5 equal parts and move 3 steps towards the left.



Example 2: Express the following rational numbers as an equivalent rational number with the given numerator or denominator.

a. $\frac{12}{15} = \frac{4}{\square}$ b. $\frac{4}{7} = \frac{8}{\square}$ c. $\frac{24}{36} = \frac{\square}{9}$

Solution: a. $\frac{12}{15} = \frac{4}{5}$ as $\frac{12 \div 3}{15 \div 3} = \frac{4}{5}$ b. $\frac{4}{7} = \frac{8}{14}$ as $\frac{4 \times 2}{7 \times 2} = \frac{8}{14}$ c. $\frac{24}{36} = \frac{6}{9}$ as $\frac{24 \div 4}{36 \div 4} = \frac{6}{9}$

Example 3: Convert the following rational numbers in standard form.

a. $\frac{224}{324}$ b. $\frac{38}{-95}$

Solution: a $\frac{224}{324} = \frac{224 \div 2}{324 \div 2} = \frac{112}{162} = \frac{112 \div 2}{162 \div 2} = \frac{56}{81}$

b. $\frac{38}{-95} = \frac{38 \div 19}{-95 \div 19} = \frac{2 \times (-1)}{-5 \times (-1)} = \frac{-2}{5}$
(HCF of 38 and 95 = 19)

COMPARISON OF RATIONAL NUMBERS

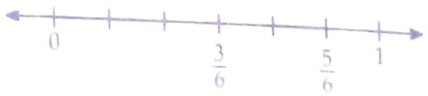
Comparison of Two Rational Numbers with the Same Denominator

If the denominators of the given rational numbers are the same, then the rational number with the greater numerator is greater, that is, the numerator of any rational number on the left of the number line is smaller than the numerator on the right.

Example 4: Compare the following rational numbers using a number line.

a. $\frac{3}{6}$ and $\frac{5}{6}$ b. 0 and $\frac{-3}{4}$

Solution: a. As $3 < 5$, $\frac{3}{6} < \frac{5}{6}$.



b. As $0 > -3$, $0 > \frac{-3}{4}$.



Comparison of Two Rational Numbers with Different Denominators

To compare two rational numbers with different denominators, follow these steps:

Step 1: Express the rational numbers with positive denominators.

Step 2: Find the LCM of both the denominators.

Step 3: Write the rational numbers as equivalent rational numbers such that their denominators are the LCM.

Step 4: Compare the numerators of the equivalent rational numbers. The rational number with the greater numerator is greater.

Example 5: Compare the following rational numbers.

a. $\frac{-3}{5}$ and $\frac{7}{-15}$ b. $\frac{5}{12}$ and $\frac{4}{16}$

Solution: **Step 1:** We can write $\frac{-3}{5}$ as $\frac{-9}{15}$ **Step 2:** LCM of 15 and 15 = 15

Step 3: Write rational numbers as equivalent rational numbers

$\frac{-3}{5} = \frac{-3 \times 3}{5 \times 3} = \frac{-9}{15}$ and $\frac{-7}{15}$

Step 4: On comparing the numerators, we get $-9 < -7$. Thus, $\frac{-9}{15} < \frac{-7}{15}$ or $\frac{-3}{5} < \frac{-7}{15}$

(ii) $\frac{5}{12}$ and $\frac{9}{16}$ LCM of 12 and 16 = 48 $\frac{5}{12} = \frac{5 \times 4}{12 \times 4} = \frac{20}{48}$ and $\frac{9}{16} = \frac{9 \times 3}{16 \times 3} = \frac{18}{48}$

On comparing the numerators, we get $20 > 18$. Thus, $\frac{20}{48} > \frac{18}{48}$ or $\frac{5}{12} > \frac{9}{16}$

COMPARISON OF TWO RATIONAL NUMBERS WITH THE SAME NUMERATOR

Any positive rational number with a smallest denominator will be greater if the numerators of the given positive rational numbers are the same.

For example, the rational numbers $\frac{-7}{9}$ and $\frac{-7}{11}$ have the same numerator. So, on comparing their denominators we get $9 < 11$. Thus, $\frac{-7}{9} > \frac{-7}{11}$

Thus, we can say if $\frac{a}{b}$ and $\frac{a}{c}$ are rational numbers and $b, c > 0$, then $\frac{a}{b} > \frac{a}{c}$ if $b < c$ and $\frac{a}{b} < \frac{a}{c}$ if $b > c$.

ORDERING OF RATIONAL NUMBERS

Example 6: Arrange the following rational numbes in ascending order:

$\frac{3}{4}, \frac{-5}{8}, \frac{8}{12}, \frac{7}{16}$ and $\frac{2}{-3}$

Solution: $\frac{2}{-3}$ can be written as $\frac{-2}{3}$

Step 1: LCM of denominators (3, 4, 8, 12 and 16) = $3 \times 2 \times 2 \times 2 \times 2 = 48$

3	3,	4,	8,	12,	16
2	1,	4,	8,	4,	16
2	1,	2,	4,	2,	8
2	1,	1,	2,	1,	4
2	1,	1,	1,	1,	2
	1,	1,	1,	1,	1

Step 2: Convert each rational number into equivalent rational numbers.

$\frac{3}{4} = \frac{3 \times 12}{4 \times 12} = \frac{36}{48}$ $\frac{-5}{8} = \frac{-5 \times 6}{8 \times 6} = \frac{-30}{48}$

$\frac{8}{12} = \frac{8 \times 4}{12 \times 4} = \frac{32}{48}$ $\frac{7}{16} = \frac{7 \times 3}{16 \times 3} = \frac{21}{48}$ $\frac{-2}{3} = \frac{-2 \times 16}{3 \times 16} = \frac{-32}{48}$

The equivalent rational numbers are $\frac{36}{48}, \frac{-30}{48}, \frac{32}{48}, \frac{21}{48}, \frac{-32}{48}$

Step 3: On comparing the numerators of equivalent rational numbers, the increasing order of the numerators is

$-32 < -30 < 21 < 32 < 36$

So, $\frac{-32}{48} < \frac{-30}{48} < \frac{21}{48} < \frac{32}{48} < \frac{36}{48}$ or $\frac{-2}{3} < \frac{-5}{8} < \frac{7}{16} < \frac{8}{12} < \frac{3}{4}$

Thus, the ascending order is $\frac{2}{-3}, \frac{-5}{8}, \frac{7}{16}, \frac{8}{12}, \frac{3}{4}$

EXERCISE 1(A)

1. Identify the rational numbers which are in standard form. Also, convert other rational numbers in standard form.

(a) $\frac{3}{-7}$ (b) $\frac{12}{14}$ (c) $\frac{-5}{0}$

2. Fill in the boxes.

(a) $\frac{-3}{16} = \frac{\square}{\square} = \frac{\square}{128}$ (b) $\frac{2}{\square} = \frac{\square}{-14} = \frac{\square}{56}$ (c) $\frac{3}{5} = \frac{\square}{\square} = \frac{\square}{\square}$

3. Convert the following rational numbers with denominator 24.

a. $\frac{7}{2}$ b. $\frac{4}{-6}$ c. $\frac{-9}{12}$ d. $\frac{32}{-48}$

4. Express the following as rational numbers with positive denominators.

a. $\frac{-18}{-21}$ b. $\frac{25}{-42}$ c. $\frac{-5}{-6}$ d. $\frac{-37}{-45}$

5. Represent the following rational numbers on the number line.

a. $\frac{5}{8}$ b. $\frac{-4}{6}$ c. $\frac{3}{-7}$ d. $\frac{6}{5}$

6. Find between which two integers the given rational numbers lie. Also, find their actual position on the number line.

a. $\frac{5}{3}$ b. $\frac{-8}{3}$ c. $\frac{13}{7}$ d. $\frac{-29}{9}$

7. Which among the following rational numbers is greater?

a. $\frac{8}{13}$ and $\frac{5}{13}$ b. $\frac{7}{14}$ and $\frac{7}{18}$ c. $\frac{-17}{51}$ and $\frac{-18}{72}$ d. $\frac{27}{56}$ and $\frac{34}{72}$

8. Fill in the blanks with <, > or = sign.

a. $\frac{-7}{9}$ ____ $\frac{-8}{9}$ b. $\frac{9}{15}$ ____ $\frac{9}{21}$ c. $\frac{14}{36}$ ____ $\frac{48}{54}$ d. $\frac{40}{75}$ ____ $\frac{50}{95}$

e. $\frac{68}{156}$ ____ $\frac{17}{26}$ f. $\frac{-15}{21}$ ____ $\frac{-20}{42}$ g. $\frac{-4}{9}$ ____ $\frac{-7}{4}$ h $\frac{-26}{30}$ ____ $\frac{-52}{60}$

9. Arrange the following rational numbers in ascending order.

a. $\frac{9}{12}, \frac{-7}{8}, \frac{17}{16}, \frac{15}{24}, \frac{1}{-4}$ b. $\frac{-6}{10}, \frac{1}{5}, \frac{-3}{15}, \frac{12}{20}, \frac{-16}{25}$ c. $\frac{-2}{3}, \frac{3}{8}, \frac{1}{7}, \frac{3}{5}, \frac{6}{9}$ d. $\frac{5}{6}, \frac{-5}{12}, \frac{-7}{20}, \frac{3}{-8}$

10. Find the two rational numbers whose absolute value is $\frac{6}{7}$.

ADDITION OF RATIONAL NUMBERS

When Denominators Are the Same

When denominators of given rational numbers are the same, add their numerators only.

Example 7: Add $\frac{7}{9} + \frac{1}{9}$.

Solution: $\frac{7}{9} + \frac{1}{9} = \frac{7+1}{9} = \frac{8}{9}$

Example 8: Add $\frac{12}{13} + \frac{5}{-13}$.

Solution $\frac{12}{13} + \frac{5}{-13} = \frac{12}{13} + \frac{(-5)}{13} = \frac{12+(-5)}{13} = \frac{7}{13}$

We compare rational numbers by making their denominators the same. Similarly, while adding rational numbers we first make their denominators the same and then add the numerators to get the answer. That is, we convert the rational numbers into equivalent rational numbers with a common denominator and add the numerators of the equivalent rational numbers. There are three possibilities when we add rational numbers with different denominators.

When Denominators are Different

When One Denominator is a Multiple/factor of the Other

Example 9: Add $\frac{11}{30} + \frac{4}{6}$.

Solution: We have denominators as 30 and 6.

30 is a multiple of 6 or 6 is one of the factors of 30. So, LCM of 6 and 30 = 30

$\Rightarrow \frac{4}{6} = \frac{4 \times 5}{6 \times 5} = \frac{20}{30}$ Thus, $\frac{11}{30} + \frac{20}{30} = \frac{11+20}{30} = \frac{31}{30} = 1\frac{1}{30}$

Example 10: Add $\frac{-3}{5} + \frac{-23}{45}$

Solution We have denominators as 5 and 45

45 is a multiple of 5 or 5 is one of the factors of 45. So, LCM of 5 and 45 = 45

$\Rightarrow \frac{-3 \times 9}{5 \times 9} = \frac{-27}{45}$ Thus $\frac{-3}{5} + \frac{-23}{45} = \frac{-27}{45} + \frac{-23}{45} = \frac{-27+(-23)}{45} = \frac{-50}{45} = \frac{-10}{9} = -1\frac{1}{9}$

When Denominator are Co-prime

Example 11: Add $\frac{-9}{13} + \frac{-6}{14}$

Solution: We have denominators as 13 and 14

13 and 14 are co-prime numbers

So, LCM of 13 and 14 = $13 \times 14 = 182$

$\Rightarrow \frac{-9 \times 14}{13 \times 14} = \frac{-126}{182}$ and $\frac{-6 \times 13}{14 \times 13} = \frac{-78}{182}$

Thus, $\frac{-9}{13} + \frac{-6}{14} = \frac{-126}{182} + \frac{-78}{182} = \frac{-126 + (-78)}{182} = \frac{-204}{182} = \frac{-102}{91} = -1\frac{11}{91}$

When Denominators have Common Factors

Example 12: Add $\frac{29}{45} + \frac{42}{75}$

Solution: We have denominators as 45 and 75.

$45 = 3 \times 3 \times 5$ and $75 = 5 \times 5 \times 3$

LCM = $3 \times 3 \times 5 \times 5 = 225$

$\Rightarrow \frac{29 \times 5}{45 \times 5} = \frac{145}{225}$ and $\frac{42 \times 3}{75 \times 3} = \frac{126}{225}$

Thus, $\frac{29}{45} + \frac{42}{75} = \frac{145}{225} + \frac{126}{225} = \frac{145 + 126}{225} = \frac{271}{225} = 1\frac{46}{225}$

Example 13: **Verify: $|x + y| \leq |x| + |y|$ for:**

a. $x = \frac{2}{3}$ and $y = \frac{-5}{7}$ b. $x = \frac{8}{5}$ and $y = \frac{-4}{9}$

Solution: a $|x + y| = \left|\frac{2}{3} + \left(\frac{-5}{7}\right)\right| = \left|\frac{2 \times 7 - 3 \times 5}{21}\right| = \left|\frac{14 - 15}{21}\right| = \left|\frac{-1}{21}\right| = \frac{1}{21}$

and $|x| + |y| = \left|\frac{2}{3}\right| + \left|\frac{-5}{7}\right| = \frac{2}{3} + \frac{5}{7} = \frac{2 \times 7 + 3 \times 5}{21} = \frac{14 + 15}{21} = \frac{29}{21} = 1\frac{8}{21}$

Since $\frac{1}{21} < 1\frac{8}{21}$, $|x + y| \leq |x| + |y|$.

b $|x + y| = \left|\frac{8}{5} + \left(\frac{-4}{9}\right)\right| = \left|\frac{8}{5} - \frac{4}{9}\right| = \left|\frac{8 \times 9 - 5 \times 4}{45}\right| = \frac{72 - 20}{45} = \frac{52}{45} = 1\frac{7}{45}$

and $|x| + |y| = \left|\frac{8}{5}\right| + \left|\frac{-4}{9}\right| = \frac{8}{5} + \frac{4}{9} = \frac{8 \times 9 + 5 \times 4}{45} = \frac{72 + 20}{45} = \frac{92}{45} = 2\frac{2}{45}$

Since $1\frac{7}{45} \leq 2\frac{2}{45}$, $|x + y| \leq |x| + |y|$.

PROPERTIES OF ADDITION OF RATIONAL NUMBERS

Closure Property: The sum of two rational numbers is also a rational number. In general, if $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} + \frac{c}{d}$ is also a rational number.

Example: a. $\frac{3}{8} + \frac{-5}{7} = \frac{3 \times 7 + (-5 \times 8)}{56} = \frac{21 - 40}{56} = -\frac{19}{56}$, which is a rational number.

b $\frac{-3}{8} + \frac{-4}{5} = \frac{(-3) \times 5 + (-4) \times 8}{8 \times 5} = \frac{-15 - 32}{40} = \frac{-47}{40}$, which is a rational number.

Thus, **rational numbers** are closed under addition.

Commutative Property: The sum of two rational numbers remains the same even, if the order in which the are added is changed. In general, if $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$.

Example: a $\frac{-2}{3} + \frac{5}{7} = \frac{(-2) \times 7 + 3 \times 5}{21} = \frac{-14 + 15}{21} = \frac{1}{21}$

and $\frac{5}{7} + \frac{(-2)}{3} = \frac{5 \times 3 + 7 \times (-2)}{21} = \frac{15 - 14}{21} = \frac{1}{21}$

Thus, $\frac{-2}{3} + \frac{5}{7} = \frac{5}{7} + \frac{-2}{3}$.

b. $\frac{-6}{5} + \frac{(-8)}{3} = \frac{(-6) \times 3 + 5 \times (-8)}{5 \times 3} = \frac{-18 - 40}{15} = \frac{-58}{15}$

and $\frac{-8}{3} + \left(\frac{-6}{5}\right) = \frac{(-8) \times 5 + 3 \times (-6)}{3 \times 5} = \frac{-40 - 18}{15} = \frac{-58}{15}$

Thus, $\frac{-6}{5} + \left(\frac{-8}{3}\right) = \frac{-8}{3} + \left(\frac{-6}{5}\right)$.

Thus addition is commutative for rational numbers.

Associative Property: The sum of three or more rational numbers remains the same even if the order in which they are grouped is changed. In general, if $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are three rational numbers, then $\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right)$.

Let us consider three rational numbers $\frac{3}{2}$, $\frac{-2}{3}$ and $\frac{1}{5}$.

$$\left|\frac{3}{2} + \frac{(-2)}{3}\right| + \frac{1}{5} = \left|\frac{3 \times 3 + 2 \times (-2)}{6}\right| + \frac{1}{5} = \left(\frac{9 - 4}{6}\right) + \frac{1}{5} = \frac{5}{6} + \frac{1}{5} = \frac{5 \times 5 + 1 \times 6}{30} = \frac{25 + 6}{30} = \frac{31}{30}$$

Now, let us group the numbers differently.

$$\frac{3}{2} + \left|\left(\frac{-2}{3}\right) + \frac{1}{5}\right| = \frac{3}{2} + \left|\frac{(-2) \times 5 + 3 \times 1}{15}\right| = \frac{3}{2} + \left|\frac{-10 + 3}{15}\right| = \frac{3}{2} + \left(\frac{-7}{15}\right) = \frac{3 \times 15 + 2 \times (-7)}{30} = \frac{45 - 14}{30} = \frac{31}{30}$$

$$\therefore \left|\frac{3}{2} + \left(\frac{-2}{3}\right)\right| + \frac{1}{5} = \frac{3}{2} + \left|\left(\frac{-2}{3}\right) + \frac{1}{5}\right|$$

Thus, addition is associative for rational numbers.

Additive Identity: The sum of any rational number and 0, is the rational number itself. In general, if $\frac{a}{b}$ is any rational number, then $\frac{a}{b} + 0 = 0 + \frac{a}{b} = \frac{a}{b}$. Here, 0 is called the additive identity or identity element of rational numbers.

Example: a. $\frac{5}{7} + 0 = \frac{5}{7} + \frac{0}{7} = \frac{5 + 0}{7} = \frac{5}{7}$ b. $0 + \left(\frac{-11}{12}\right) = \frac{0}{12} + \left(\frac{-11}{12}\right) = \frac{0 - 11}{12} = \frac{-11}{12}$

Additive Inverse: If $\frac{a}{b}$ is a rational numbers, then $\frac{-a}{b}$ is a rational number such that $\frac{a}{b} + \left(\frac{-a}{b}\right) = \left(\frac{-a}{b}\right) + \frac{a}{b} = 0$.

Here, $\frac{a}{b}$ and $\frac{-a}{b}$ are said to be the additive inverse or negative of each other.

Example: $\frac{4}{7} + \frac{-4}{7} = \frac{4 + (-4)}{7} = \frac{0}{7} = 0$. Thus, $\frac{4}{7}$ and $\frac{-4}{7}$ are the additive inverse of each other.

Remember

The associative property enables us to add three or more rational numbers by rearranging and grouping them in a convenient form so that the sum may be found more easily. For this, we group the given rational numbers in such a way that each group contains a pair of rational numbers with a common denominator or their denominators have a common divisor.

Example 14: Simplify: $\frac{3}{7} + \left(\frac{-6}{11}\right) + \left(\frac{-8}{21}\right) + \frac{5}{22}$

Solution $\frac{3}{7} + \left(\frac{-6}{11}\right) + \left(\frac{-8}{21}\right) + \frac{5}{22} = \left|\frac{3}{7} + \left(\frac{-8}{21}\right)\right| + \left|\left(\frac{-6}{11}\right) + \frac{5}{22}\right| = \left|\frac{3 \times 3 + (-8)}{21}\right| + \left|\frac{(-6) \times 2 + 5}{22}\right|$

$= \left(\frac{9 - 8}{21}\right) + \left(\frac{-12 + 5}{22}\right) = \frac{1}{21} + \left(\frac{-7}{22}\right) = \frac{22 - 147}{462} = \frac{-125}{462}$

EXERCISE 1(B)

1. Add the following rational numbers.

a. $\frac{3}{11} + \frac{-6}{11}$ b. $\frac{23}{41} + \frac{33}{41}$ c. $\frac{-19}{33} + \frac{21}{33}$ d. $\frac{-5}{13} + \frac{-7}{13}$

e. $\frac{-52}{64} + \frac{72}{64}$ f. $\frac{-9}{12} + \frac{1}{12}$ g. $\frac{14}{16} + \frac{-5}{24}$ h. $\frac{-5}{6} + \frac{-4}{15}$

2. Find the sum of the following rational numbers:

a. $\frac{-5}{13} + \frac{11}{26}$ b. $\frac{17}{19} + \frac{21}{57}$ c. $\frac{9}{21} + \frac{18}{27}$ d. $\frac{15}{20} + \left(\frac{-25}{35}\right)$

e. $\frac{-8}{25} + \left(\frac{-8}{75}\right)$ f. $\frac{-33}{81} + \frac{15}{18}$ g. $\frac{-1}{9} + \frac{3}{5} + \left(\frac{-5}{7}\right)$ h. $\frac{15}{27} + \left(\frac{-21}{18}\right) + \frac{24}{54}$

3. Match the following:

a. $\left(\frac{3}{5} + \frac{-4}{9}\right) + \frac{-9}{12} = \frac{7}{5} + \left(\frac{-4}{9} + \frac{9}{12}\right)$ i. Additive identity

b. $\frac{5}{6} + \frac{-8}{12} = 1\frac{1}{2}$ ii. Commutative property

c. $\frac{-5}{7} + \frac{-8}{9} = \frac{-8}{9} + \frac{-5}{7}$ iii. Additive inverse

d. $\frac{9}{13} + \frac{-9}{13} = 0$ iv. Associative property

e. $\frac{9}{13} + 0 = \frac{9}{13}$ v. Closure property

4. Fill in the blanks.

a. $\left(\frac{2}{5} + \frac{-6}{9}\right) + \frac{3}{7} = \frac{2}{5} + \left(\frac{-6}{9} + \frac{3}{7}\right)$ b. $\frac{15}{17} + \frac{-15}{17} = 0$ c. $\frac{-9}{11} + __ = \frac{-9}{11}$

d. $\frac{3}{-7} + __ = \frac{-11}{14} + \frac{-3}{7}$ e. $\frac{21}{23} + __ = __ + \frac{18}{27}$ f. $__ + \frac{-2}{9} = 0$

5. Find the additive inverse of the following:

a. $\frac{-7}{8}$ $\frac{-8}{7}$ b. $\frac{10}{11}$ $\frac{+11}{+10}$ c. $\frac{1}{1}$ d. $\frac{2}{-5} \cdot \frac{2}{-2}$

6. Simplify the following using the properties of addition.

a. $\frac{1}{7} + \left(\frac{-5}{6}\right) + \left(\frac{-13}{21}\right)$ b. $\frac{-7}{25} + \left(\frac{-17}{50}\right) + \frac{47}{100}$ c. $\frac{2}{10} + \frac{8}{5} + \frac{-1}{5}$ d. $\frac{3}{20} + \left(\frac{-2}{5}\right) + \frac{8}{10}$

7. Prove that the commutative property holds true for $\frac{-27}{56} + \frac{52}{64}$.

8. Verify $(a + b) = (b + a)$ taking $a = \frac{-3}{7}$ and $b = \frac{15}{14}$.

9. Verify $(a + b) + c = a + (b + c)$ for $a = \frac{-5}{7}$, $b = \frac{3}{5}$ and $c = \frac{7}{10}$.

10. Write the rational number which is its own additive inverse.

SUBTRACTION OF RATIONAL NUMBERS

Rational numbers are subtracted in the same way as we subtract fractions. The rational numbers are converted to equivalent rational numbers where the denominator is equal to the LCM of denominators. Then the numerators are subtracted. Let us observe the following cases of subtraction.

Case 1: **When both the rational number are positive.**

Example 15: Subtract $\frac{2}{7}$ from $\frac{5}{14}$.

Solution: $\frac{5}{14} - \frac{2}{7} = \frac{5}{14} - \frac{2 \times 2}{7 \times 2} = \frac{5}{14} - \frac{4}{14} = \frac{5-4}{14} = \frac{1}{14}$

Case 2: **When the rational number to be subtracted has a negative numerator.**

Example 16: Subtract $\frac{-4}{15}$ from $\frac{3}{5}$.

Solution: $\frac{3}{5} - \left(\frac{-4}{15}\right) = \frac{3 \times 3}{5 \times 3} - \left(\frac{-4}{15}\right) = \frac{9}{15} + \frac{4}{15} = \frac{9+4}{15} = \frac{13}{15}$

Case 3: **When the rational number from which we have to subtract has a negative numerator**

Example 17: Subtract $\frac{6}{8}$ from $\frac{-3}{12}$.

Solution: $\frac{-3}{12} - \frac{6}{8} = \frac{(-3) \times 2}{12 \times 2} = \frac{6 \times 3}{8 \times 3} = \frac{-6}{24} - \frac{18}{24} = \frac{(-6) - 18}{24} = \frac{-24}{24} = -1$

Case 4: **When both the rational numbers are negative**

Example 18: Subtract $\frac{-7}{18}$ from $\frac{-9}{15}$.

Solution: $\frac{-9}{15} - \left(\frac{-7}{18}\right) = \frac{(-9) \times 6}{15 \times 6} - \frac{(-7) \times 5}{18 \times 5} = \frac{-54}{90} - \left(\frac{-35}{90}\right) = \frac{-54 + 35}{90} = \frac{-19}{90}$.

PROPERTIES OF SUBTRACTION OF RATIONAL NUMBERS

Closure Property: If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} - \frac{c}{d}$ is also a rational number.

Example: a. $\frac{-5}{7} - \frac{2}{3} = \frac{(-5) \times 3 - 2 \times 7}{7 \times 3} = \frac{-15 - 14}{21} = \frac{-29}{21}$, which is a rational number.

b. $\frac{3}{7} - \left(\frac{-8}{5}\right) = \frac{3}{7} + \frac{8}{5} = \frac{3 \times 5 + 7 \times 8}{7 \times 5} = \frac{15 + 56}{35} = \frac{71}{35}$, which is a rational number.

Thus, **rational numbers are closed under subtraction**.

Commutative Property: The communative property does not hold true for rational numbers.

Example: $\frac{2}{3} - \frac{5}{4} = \frac{2 \times 4 - 3 \times 5}{3 \times 4} = \frac{8 - 15}{12} = \frac{-7}{12}$ and $\frac{5}{4} - \frac{2}{3} = \frac{3 \times 5 - 4 \times 2}{4 \times 3} = \frac{15 - 8}{12} = \frac{7}{12}$

So, $\frac{2}{3} - \frac{5}{4} \neq \frac{5}{4} - \frac{2}{3}$.

Thus, **subtraction is not commutative for rational numbers.**

Associative Property: The associative property does not hold true for rational numbers.

Example: $\frac{-2}{3} - \left[\frac{-4}{5} - \frac{1}{2}\right] = \frac{-2}{3} - \left[\frac{-4 \times 2 - 1 \times 5}{10}\right] = \frac{-2}{3} - \left[\frac{-8 - 5}{10}\right] = \frac{-2}{3} - \left(\frac{-13}{10}\right) = \frac{-2}{3} + \frac{13}{10}$

$= \frac{-2 \times 10 + 3 \times 13}{30} = \frac{-20 + 39}{30} = \frac{19}{30}$

And, $\left[\frac{-2}{3} - \left(\frac{-4}{5}\right)\right] - \frac{1}{2} = \left[\frac{-2}{3} + \frac{4}{5}\right] - \frac{1}{2} = \left[\frac{-2 \times 5 + 3 \times 4}{15}\right] - \frac{1}{2} = \left[\frac{-10 + 12}{15}\right] - \frac{1}{2} = \frac{2}{15} - \frac{1}{2}$

$= \frac{2 \times 2 - 15 \times 1}{30} = \frac{4 - 15}{30} = \frac{-11}{30}$

So, $\frac{-2}{3} - \left[\frac{-4}{5} - \frac{1}{2}\right] \neq \left[\frac{-2}{3} - \left(\frac{-4}{5}\right)\right] - \frac{1}{2}$.

Thus, the subtraction of rational numbers is not associative.

Property of Zero: The result of subtracting zero from a rational number is the rational number itself.
For all rational numbers, $\frac{a}{b} - 0 = \frac{a}{b}$; but $0 - \frac{a}{b} \neq \frac{a}{b}$

Thus, **no identity exists for subtraction**.

Example: $\frac{3}{7} - 0 = \frac{3 - 0}{7} = \frac{3}{7}$ and $0 - \frac{3}{7} = \frac{0 - 3}{7}$

Thus, $\frac{3}{7} - 0 \neq 0 - \frac{3}{7}$